Malaysia Industrial Mathematical Modelling Challenge 2026 (MIMMC2026)

Solution Report

Problem Title:

Dynamic Pricing for

Agricultural Machinery Rental Systems

Team ID Number:

MIMMC_23

ABSTRACT

This report presents a mathematical modelling approach for developing a fair and sustainable dynamic pricing strategy for agricultural machinery rental services in chili farming. The study considers the interactions between three key stakeholders: farmers, service providers, and market demand. The proposed model integrates demand behaviour, farmer profitability, and operational cost structures to evaluate how rental pricing affects service adoption and financial sustainability. Mechanisation is shown to improve farming efficiency through higher crop productivity and reduced labour dependency, while also providing profitable returns for service providers.

An optimal rental pricing strategy is identified by balancing affordability for farmers, profitability for providers, and stable market demand. The model also examines the feedback relationship between pricing, adoption rate, and long-term system sustainability. Sensitivity analysis is conducted to evaluate the impact of factors such as chili market prices, operational costs, demand sensitivity, and service capacity. The findings demonstrate that an appropriate pricing strategy can create a stable equilibrium where all stakeholders benefit simultaneously. The proposed framework provides a practical and data-driven approach that may support sustainable agricultural mechanisation and rental service planning in the future.

SOLUTION

Task 1: Integrated Mathematical Model

This model integrates three interdependent components: demand behaviour, farmer economics, and service provider cost-profitability. Together they form a dynamic pricing system for agricultural machinery rental.

1(i) Demand Component (Adoption Behaviour)

The demand model uses a linear price-response function that captures how rental price influences the total area of land service.

Model Formulation

General form: $Q = Q_0 - \left(\beta \times Q_0\right)\left(P - P_0\right)$

Where:

$\beta = 0.0015$ (price sensitivity factor)

$Q_0 = 80$ *hectare* (base area serviced at reference price)

$P_0 = RM5200/$ *hectare* (reference adoption price)

Q1: How does rental price affect total area serviced?

Substituting the values given:

$$\begin{aligned} Q &= 80 - (0.0015 \times 80)(P - 5200) \\ &= 704 - 0.12P \end{aligned}$$

The slope coefficient 0.12 means for every RM1 increase in rental price, the total area serviced decreases by 0.12 hectares. The price sensitivity factor $\beta = 0.0015$ quantifies how strongly farmers respond to price changes per unit of base demand.

Q2: How does farmer adoption vary with affordability?

The demand function $Q = 704 - 0.12P$ directly captures adoption behavior. As price rises above the reference level of RM5,200/hectare, adoption rate decreases proportionally. For every RM100 increase beyond P_0, adoption falls by 12 hectares.

Q3: Maximum potential demand under optimal conditions?

At the reference adoption price $P = P_0 = RM\ 5,200/\ hectare$:

$$Q = 80 - 0.12(5200 - 5200)$$
$$= 80\ hectare$$

The maximum realistic demand is $Q = 80\ hectare$, constrained by system service capacity. While the mathematical model yields $Q = 704\ hectare$ at $P = 0$, this is physically infeasible. The system capacity sets the upper bound at 80 *hectare*.

Q4: Critical price threshold

Setting $Q = 0$ to find the price at which all farmers abandon the service:

$$0 = 704 - 0.12P$$
$$P = \frac{704}{0.12} = RM5866.67/\ hectare$$

Critical Threshold:

P = RM5,866.67/hectare (beyond this price, demand collapses entirely to zero)

1(ii) Farmer Component (Economic Benefit and Decision-Making)

Q1: Yield increases from mechanization

Method	Yield (kg/ hectare)	Difference
Mechanized	10,200 kg/hectare	
Traditional	9,500 kg/ hectare	**+700 kg/hectare**

$$\triangle Y = 10200 - 9500 = 700\ kg/hectare$$

$$\frac{700}{9500} \times 100 = 7.37\%$$

For every 1 hectare of land farmed using mechanisation, crop yield increases by 700 kg, representing a 7.37% improvement over traditional farming methods.

Q2: Labour cost savings

Method	Labour Cost (MYR/hectare)	Savings
Mechanised	RM800	
Traditional	RM2,200	**RM1,400 saved**

$$\triangle L = RM\ 2200 - RM\ 800 = RM\ 1400/hectare$$

$$\frac{1400}{2200} \times 100 = 63.64\%$$

For every 1 hectare of land farmed using mechanisation, labour cost decreases by RM1400, representing a 63.64% reduction compared to traditional farming methods.

Q3: Revenue from chili production

Method	Yield	Chili Price	Revenue
Mechanised	10,200 kg/ hectare	RM6.00/kg	RM61,200/ hectare
Traditional	9,500 kg/ hectare	RM6.00/kg	RM57,000/ hectare
Difference	+700 kg/ hectare	-	**+RM4,200/** hectare

$$Revenue\ gain = 700\ kg/hectare \times RM6.00/kg = RM4200/hectare$$

$$\frac{4200}{57000} \times 100 = 7.37\%$$

For every 1 hectare of land farmed using mechanisation, revenue increases by RM4,200 representing a 7.37% improvement over traditional farming methods.

Q4: Net financial gain/loss after rental cost

Net profit, π formula for each method:

$$\pi_{mech} = Revenue(mech) - Labour(mech) - RentalPrice$$
$$= RM61200 - RM800 - P$$
$$= RM\ 60400 - P$$

$$\pi_{trad} = Revenue(trad) - Labour(trad)$$
$$= RM\ 57000 - RM\ 2200$$
$$= RM\ 54800/hectare$$

At reference price $P = RM5{,}200/hectare$:

$$\pi_{mech} = RM\ 60400 - RM5200 = RM55200/hectare$$
$$\pi_{trad} = RM54800/hectare$$
$$Advantage = RM55200 - RM\ 54\ 800 = RM400/hectare$$

Farmer breakeven rental price (mechanization = traditional):

$$RM60400 - P = RM\ 54800$$
$$P_{max}(farmer) = RM5600/\ hectare$$

Farmer Conclusion:

Mechanization is financially beneficial for any rental price below RM5,600/ hectare. At the reference price of RM5,200/ hectare, farmers gain an additional RM400/ hectare profit compared to traditional methods.

1(iii) Service Provider Component (Cost and Profitability)

<u>Capital Investment and Depreciation</u>

$$\begin{aligned} Annual\ Depreciation\ (straightline) &= (Cost - Salvage)\ /\ Lifespan \\ &= (RM85000 - RM15000)\ /\ 8 \\ &= RM70000\ /\ 8 \\ &= RM8750/\ year \end{aligned}$$

Depreciation represents the annual hidden cost of asset value decline. Although no cash leaves the business annually, it must be recovered through rental pricing to fund future machinery replacement.

Annual Fixed Costs

$$FC = Annual\ Maintenance + Annual\ Depreciation$$
$$\begin{aligned} FC &= RM2,550 + RM8,750 \\ &= RM11,300/\ year \end{aligned}$$

Variable Operational Costs per Hectare

$$VC = Fuel\ Cost + Operator\ Salary$$
$$\begin{aligned} VC &= RM120 + RM150 \\ &= RM270/\ hectare \end{aligned}$$

Variable costs scale directly with usage. Higher demand means higher total variable expenditure.

Wear-Related Costs from Intensive Usage

Intensive usage accelerates machinery wear beyond the standard depreciation schedule. This is modelled as an additional wear factor k applied to variable costs:

$$Wearcost = k \times VC/hectare$$
$$where,\ k = 0.05(5\%\ wearf actor)$$
$$= 0.05 \times RM270$$
$$= RM13.50/hectare$$
$$Total\ VC(incl.wear) = RM270 + RM13.50$$
$$= RM283.50/hectare$$

The 5% wear factor is a conservative estimate reflecting additional servicing and part replacements under high utilisation. This can be revised based on field data.

Total Cost and Required Revenue

Total Annual Cost at demand Q:

$$TC(Q) = FC + VC\ total \times Q$$
$$TC(Q) = 11,300 + 283.50 \times Q$$

At reference demand Q = 80 hectare:

$$TC = 11,300 + 283.50 \times 80$$
$$TC = 11,300 + 22,680$$
$$= RM33,980/year$$

Desired Profit Margin

The provider targets a 25% profit margin on revenue (not on cost). Using the revenue-based margin formula:

$$TR = TC / (1 - margin) = RM33,980 / (1 - 0.25) = RM45,306.67/year$$

Minimum rental price to sustain 25% profit:

$$P_{min} = TR/Q = RM45,306.67/80 = RM566.33/hectare$$

Cost Component	Value	Type
Annual Depreciation	RM8,750/year	Fixed
Annual Maintenance	RM2,550/year	Fixed
Fuel Cost	RM120/ hectare	Variable
Operator Salary	RM150/ hectare	Variable
Wear-Related Cost	RM13.50/ hectare	Variable
Total Fixed Cost	**RM11,300/year**	
Total Variable Cost	**RM283.50/ hectare**	
Total Annual Cost (Q=80)	**RM33,980/year**	
Minimum Rental Price	**RM566.33/ hectare**	+25% margin

Task 2: Optimal and Fair Pricing Strategy

An optimal rental price must simultaneously satisfy all three stakeholders. The following boundaries define the feasible pricing range.

Boundary	Price (MYR/hectare)	Condition	Stakeholder
Lower Bound	RM566.33	Provider covers costs + 25% profit	Service Provider
Upper Bound (farmer)	RM5,600.00	Farmer still benefits vs. traditional	Farmer
Upper Bound (demand)	RM5,866.67	Demand does not collapse to zero	Demand
Reference / Optimal	**RM5,200.00**	**All three conditions satisfied**	**All Parties**

Justification of P = 5200/hectare as Optimal Price:

Condition (a): Farmers derive measurable economic benefit:

$$\pi_{mech} = RM60,400 - RM5,200$$
$$= RM55,200/\,hectare$$
$$\pi_{trad} = RM54,800/\,hectare$$
$$Farmer\ gain = RM55,200 - RM54,800$$
$$= RM400/\,hectare$$
$$\therefore\ proven$$

Condition (b): Service provider achieves profitability:

$$TR = RM5,200 \times 80$$
$$= RM416,000/\,year$$
$$TC = RM33,980/\,year$$
$$Provider\ prof\,it = RM382,020/\,year$$
$$\therefore\ proven$$

Condition (c): Demand remains within operational bounds:

$$\begin{aligned} Q &= 704 - 0.12(5200) \\ &= 80\ hectare \\ &= system\ capacity \\ &\therefore\ proven \end{aligned}$$

Optimal Price Range:

RM566.33/hectare to RM5,600/hectare. The reference price of RM5,200/hectare is recommended as it aligns with the market adoption price, satisfies all three stakeholders, and keeps demand at fully system capacity.

Task 3: System Interactions and Feedback Dynamics

The three components are not independent; they form a closed feedback loop. A change in rental price propagates through the entire system.

3(a) Rental price influences farmer adoption and total serviced area

The demand function directly links price to serviced area:

$$Q(P) = 704 - 0.12P$$

As price increases, fewer farmers adopt the service, reducing Q. This is the primary price signal in the system.

3(b) Total demand drives operational costs and ultimately determines the service provider's profit

Provider profit depends on total demand through:

$$TC(Q) = 11,300 + 283.50 \times Q$$
$$TR(P,Q) = P \times Q(P)$$
$$\pi_{provider} = TR - TC$$
$$= P \times (704 - 0.12P) - [11,300 + 283.50(704 - 0.12P]$$

The following table quantifies how the system responds across different rental prices:

Price (MYR/hectare)	Q (hectare)	TC (MYR/yr)	TR (MYR/yr)	Provider Profit	Farm Net Profit
RM566	636.1	RM192,023	RM360,085	RM168,062	RM59,834
RM2,000	464.0	RM142,844	RM928,000	RM785,156	RM58,400
RM4,000	224.0	RM74,804	RM896,000	RM821,196	RM56,400
RM5,200*	**80.0**	**RM33,980**	**RM416,000**	**RM382,020**	**RM55,200**
RM5,500	44.0	RM23,774	RM242,000	RM218,226	RM54,900
RM5,867	0.0	RM11,300	RM0	-RM11,300	RM54,533

*Recommended optimal price

3(c) Farmer economic outcomes feedback into future adoption decisions, creating a dynamic feedback loop.

Farmer net profit feeds back into future demand decisions, creating a dynamic loop:

Step 1: Provider sets rental price P

Step 2: Demand responds: $Q = 704 - 0.12P$

Step 3: Provider incurs costs $TC = 11,300 + 283.50Q$

Earns $TR = P \times Q$

Step 4: Farmer computes net profit: $\pi_{\text{mech}} = RM60,400 - P$

Step 5: If $\pi_{\text{mech}} > \pi_{\text{trad}}$, adoption grows in the next season, otherwise it contracts

Step 6: Shift in adoption changes Q, which affects TC which may pressure the provider to revise P

This feedback loop is stabilising at $P = RM5{,}200/hectare$ because farmer profit exceeds traditional (positive adoption signal), demand equals system capacity (no excess or shortage), and the provider earns well above minimum required revenue.

Task 4: Sensitivity Analysis

Sensitivity analysis examines how the system responds when individual parameters deviate from baseline values. All analyses are conducted at the reference price $P = RM5{,}200/\ hectare$ and $Q = 80$ hectare unless stated otherwise.

4(a) Chili Market Price Fluctuation

Changes in chili market price directly affect farmer revenue and the maximum rental price farmers are willing to pay.

Farmer breakeven formula:

$$P_{max}(farmer) = (Y_{mech} - Y_{trad}) \times c_{chili} + (L_{trad} - L_{mech})$$
$$= 700 \times c_{chili} + 1,400$$

Chili Price	Mech Revenue	Farmer Net Profit (at P=5,200)	Breakeven Rental P	Risk Level
RM4.00/kg	RM40,800/ hectare	RM34,800/ hectare	RM4,200/ hectare	**High Risk**
RM5.00/kg	RM51,000/ hectare	RM45,000/ hectare	RM4,900/ hectare	Moderate
RM6.00/kg*	**RM61,200/ hectare**	**RM55,200/ hectare**	**RM5,600/ hectare**	**Stable**
RM7.00/kg	RM71,400/ hectare	RM65,400/ hectare	RM6,300/ hectare	Low Risk
RM8.00/kg	RM81,600/ hectare	RM75,600/ hectare	RM7,000/ hectare	Very Low

*Baseline value

If chili prices fall to RM4.00/kg, the farmer breakeven price drops to RM4,200/ hectare, which is below the reference rental price of RM5,200/ hectare. This signals adoption collapse risk. The provider should monitor chili market prices and consider dynamic pricing adjustments when chili prices fall below RM5.00/kg.

4(b) Price Sensitivity Factor (β)

A higher β means farmers react more strongly to price changes, causing demand to fall faster with each price increase.

β Value	**Slope (hectare /MYR)**	**Critical Price**	**Q at P = 5,200**	**Market Sensitivity**
0.0005	0.040	RM7,200/hectare	80 hectares	Low
0.0010	0.080	RM6,200/hectare	80 hectares	Moderate
0.0015*	**0.120**	**RM5,867/ hectare**	**80 hectares**	**Baseline**
0.0020	0.160	RM5,700/hectare	80 hectares	High
0.0030	0.240	RM5,533/hectare	80 hectares	Very High

*Baseline value

Demand at $P = P_0 = RM5{,}200$ remains $Q = 80$ hectare regardless of β, since the reference price always returns to the base demand. However, a higher β drastically narrows the safe pricing window which at $\beta = 0.003$, the critical threshold is only RM5,533/hectare, leaving very little room to raise prices above the reference without losing all demand. The provider must be cautious about price increases in markets with high β.

4(c) Operational Cost Escalation

Fuel, labour, and maintenance cost increases raise the minimum rental price required for profitability.

Cost Level	VC/hectare (MYR)	TC at Q=80	P_{min} Required	Prov. Profit at P=5,200	Status
-20%	RM226.80	RM29,444	RM490.73	RM386,556	Safe
-10%	RM255.15	RM31,712	RM528.53	RM384,288	Safe
Baseline*	**RM283.50**	**RM33,980**	**RM566.33**	**RM382,020**	**Stable**
+20%	RM340.20	RM38,516	RM641.93	RM377,484	Manageable
+50%	RM425.25	RM45,320	RM755.33	RM370,680	Manageable

*Baseline value

Even a 50% escalation in operational costs only raises the minimum viable rental price to RM755.33/hectare which is far below the current reference price of RM5,200/hectare. This indicates that the system is highly resilient to operational cost shocks. The provider's large profit buffer (RM382,020 at baseline) provides substantial protection against cost increases.

4(d) Maximum Service Capacity

If the system's service capacity is reduced (e.g., machinery breakdown, fewer units) or expanded, the economic outcomes shift accordingly.

Q_{max}(hectare)	Q Served	$TC\ (MYR/yr)$	P_{min} Required	Prov. Profit at P = 5,200
40 hectares	40 hectares	RM22,640	RM754.67/hectare	RM185,360
60 hectares	60 hectares	RM28,310	RM629.11/hectare	RM283,690
80 hectares*	**80 hectares**	**RM33,980**	**RM566.33/hectare**	**RM382,020**
≥100 hectares	80 hectares	RM33,980	RM566.33/hectare	RM382,020

*Baseline + Demand is capped at Q_0 = 80 hectares at reference price; extra capacity is idle unless price is reduced.

Reducing capacity to 40 hectares raises the minimum viable price to RM754.67/hectare and halves provider profit. Expanding capacity beyond 80 hectare yields no additional benefit at the current reference price, since market demand at $P = RM5{,}200$ is already at capacity. Capacity expansion only becomes valuable if the provider reduces price to attract additional demand from the wider market.

SUMMARY AND CONCLUSION

Parameter	Value	Description	Stakeholder
Optimal Rental Price	**RM5,200/hectare**	Reference adoption price	All
Provider P_{min}	RM566.33/hectare	Break-even + 25% margin	Provider
Farmer P_{max}	RM5,600/hectare	Mechanisation stays beneficial	Farmer
Demand P_{max}	RM5,866.67/hectare	Above this, demand = 0	Demand
Provider Annual Profit	RM382,020	At P = RM5,200, Q = 80 hectare	Provider
Farmer Net Advantage	RM400/hectare	vs. traditional farming	Farmer

The integrated model demonstrates that agricultural machinery rental at RM5,200 per hectare represents a stable and fair equilibrium. The service provider earns substantial profit, farmers gain measurably over traditional methods, and demand precisely matches system capacity. The feedback dynamics further reinforce this equilibrium: positive farmer outcomes drive continued adoption, which sustains provider revenue. The primary systemic risk identified through sensitivity analysis is a decline in chili market price below RM5.00/kg, which would erode farmer willingness to pay and may trigger adoption contraction.

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